Degree and square-root energy on the Boolean cube: equivalent positive estimates and a signed obstruction

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October 16, 2026

Abstract

For nonnegative Boolean-cube polynomials, we show that any universal square-root energy bound which vanishes with the degree-to-dimension ratio implies a linear degree bound. The same assertion is equivalent to approximation, in relative L^1 error, by sums of squares of polynomials of degree proportional to the original degree. One fixed relative approximation error strictly below one suffices. The proof uses exact linear-code embeddings and tensor powers. For arbitrary signed polynomials the analogous square-root statement is false, even with a global unit absolute margin and with all Fourier coefficients on one level. The signed obstruction persists at any fixed prediction accuracy below one for full parity, without a bound on the predictor's source cost. These are auxiliary analytic results. The positive inequality itself, and the intended natural-scale posterior sample complexity bound for sparse parity, remain open.

1 Definitions and the positive equivalence

All expectations on $\{-1,1\}^k$ use the uniform probability measure. Write $\chi_S(x) = \prod_{i\in S} x_i$ and

$$\Delta\chi_S = |S|\chi_S, \qquad \mathcal{E}(q) = \mathbb{E}q\Delta q = \sum_S |S|\widehat{q}(S)^2.$$

Degree always means exact multilinear Walsh degree. For $P \geq 0$, the quantity considered below is $\mathcal{E}(\sqrt{P})$, and $\|\sqrt{P}\|_2^2 = \mathbb{E}P$. Zero polynomials and constants are harmless and may be omitted from statements involving their degree in a denominator.

Theorem 1. *The following universal assertions are equivalent. Each is quantified over every dimension k and every nonzero nonnegative polynomial P on the k-cube of positive degree r.*

(i) There is a finite absolute constant C such that

$$\mathcal{E}(\sqrt{P}) \leq Cr\mathbb{E}P.$$

(ii) There is $\omega : [0,1] \to [0,\infty)$ with $\omega(t) \to 0$ as $t \downarrow 0$ such that

$$\mathcal{E}(\sqrt{P}) \leq k\omega(r/k)\mathbb{E}P.$$

(iii) For each $\varepsilon > 0$ there is a finite C_ε such that

$$\sum_{|S|>C_\varepsilon r} \widehat{\sqrt{P}}(S)^2 \leq \varepsilon\mathbb{E}P.$$

(iv) For each $\varepsilon > 0$ there is a finite D_ε such that there are real polynomials Q_j with

$$\deg Q_j \leq D_\varepsilon r, \qquad \mathbb{E}\left|P - \sum_j Q_j^2\right| \leq \varepsilon\mathbb{E}P.$$

Moreover, (iii) for one fixed $\varepsilon < 1$ already implies (i). Likewise, (iv) for one fixed relative error $\varepsilon < 1$ already implies (i). If (iv) holds for all errors, its approximant can always be chosen to be a single square, with a possibly different degree constant.

The theorem is an equivalence of existence statements. It does not establish any of them or assert a sharp constant. In particular, a weak degree-fraction estimate on the full positive cone is not an easier existence target than the linear estimate.

2 An exact noise embedding

We identify cube coordinates with $\mathbb{F}_2$ bits when applying a binary linear map. This identification does not change the Walsh notation.

Lemma 2. *Let $\theta \in (0, 1/2)$ be rational. For every k there are an integer K and a surjective binary linear map $A : \mathbb{F}_2^K \to \mathbb{F}_2^k$ such that the lift $\widetilde{P}(Y) = P(AY)$ preserves the distribution of P and sends a Walsh character of degree j to a character of degree*

$$d_\theta(j) = \frac{K}{2}\big(1 - (1-2\theta)^j\big) \le K\theta j.$$

For every real function q on the original cube,

$$\frac{\mathcal{E}(q \circ A)}{K} = \frac{1}{2}\sum_S \big(1 - (1-2\theta)^{|S|}\big)\widehat{q}(S)^2. \tag{1}$$

Proof. Let μ_θ be the law on $\mathbb{F}_2^k$ with independent Bernoulli(θ) coordinates. Choose a common denominator K for all its masses, and form a k-by-K matrix with exactly $K\mu_\theta(v)$ copies of each v as columns. The zero column is allowed and merely creates an irrelevant physical variable. Every standard basis vector occurs, so the matrix has full row rank and AY is uniform when Y is uniform. The lift of χ_S has degree $K\mathbb{P}_{v\sim\mu_\theta}(v \cdot S = 1) = d_\theta(|S|)$. Distinct characters remain distinct by full row rank. Parseval's identity now gives (1). □

3 Proof of the positive equivalence

First, (i) implies (ii) and (iii), using Markov's inequality for the nonnegative Fourier masses of $\sqrt{P}$.

Suppose (ii). We may replace ω near zero by a nondecreasing envelope, which still tends to zero. Apply Lemma 2 to P. Since $\deg \widetilde{P}/K \le \theta r$, it gives

$$\frac{1}{2}\sum_S \big(1 - (1-2\theta)^{|S|}\big)\widehat{\sqrt{P}}(S)^2 \le \omega(\theta r)\mathbb{E}P. \tag{2}$$

For rational $c \in (0, 1/2)$ take $\theta = c/r$. If $|S| > r/c$, then $(1-2\theta)^{|S|} < e^{-2}$. Consequently

$$\sum_{|S|>r/c} \widehat{\sqrt{P}}(S)^2 \le \frac{2\omega(c)}{1-e^{-2}}\mathbb{E}P.$$

Letting c be a sufficiently small fixed rational proves (iii).

Next suppose (iii) holds for one fixed $\varepsilon < 1$, with constant $C_\varepsilon = C$. Normalize $\mathbb{E}P = 1$ and put $q = \sqrt{P}$. The masses $\widehat{q}(S)^2$ define a probability distribution; let J be the degree $|S|$ under this distribution. Then $\mathbb{E}J = \mathcal{E}(q)$. For the tensor power

$$P_m(x^{(1)}, \dots, x^{(m)}) = \prod_{a=1}^m P(x^{(a)}),$$

degree is mr, and the Fourier degree distribution of $\sqrt{P_m}$ is the law of $J_1 + \cdots + J_m$ for independent copies of J. Thus $\mathbb{P}(J_1 + \cdots + J_m > Cmr) \le \varepsilon$ for every m. If $\mathbb{E}J > Cr$, the weak law of large numbers makes this probability tend to one. Therefore $\mathcal{E}(\sqrt{P}) \le Cr$, proving (i).

For (iii)⇒(iv), let Q be the Fourier projection of $q = \sqrt{P}$ to degrees at most $C_\delta r$. Cauchy–Schwarz gives

$$\mathbb{E}|P - Q^2| \le \|q - Q\|_2(\|q\|_2 + \|Q\|_2) \le 2\sqrt{\delta}\,\mathbb{E}P.$$

Choose $\delta = (\varepsilon/2)^2$.

Conversely, suppose that for one fixed $\delta < 1$ there is $R = \sum_j Q_j^2$ with $\deg Q_j \le Dr$ and $\mathbb{E}|P - R| \le \delta \mathbb{E}P$. The Euclidean norm is Lipschitz, so on every edge

$$(\sqrt{R(x)} - \sqrt{R(y)})^2 \le \sum_j (Q_j(x) - Q_j(y))^2.$$

It follows that $\mathcal{E}(\sqrt{R}) \le Dr\mathbb{E}R \le Dr(1+\delta)\mathbb{E}P$. Also $\|\sqrt{P} - \sqrt{R}\|_2^2 \le \mathbb{E}|P - R|$. Hence

$$\frac{\|\Pi_{>Cr}\sqrt{P}\|_2}{\sqrt{\mathbb{E}P}} \le \sqrt{\delta} + \sqrt{\frac{D(1+\delta)}{C}}.$$

For a sufficiently large fixed C, the right side is strictly below one. We have obtained a fixed version of (iii), which already implies (i). This proves all assertions of Theorem 1.

There is also a direct constant conversion. In (2), apply the embedding to P_m with $\theta = c/(mr)$. The left side, after dividing by $\mathbb{E}P_m$, tends to

$$\frac{1}{2}\left[1 - \exp\left(-\frac{2c\mathcal{E}(\sqrt{P})}{r\mathbb{E}P}\right)\right].$$

Thus whenever $\omega(c) < 1/2$,

$$\frac{\mathcal{E}(\sqrt{P})}{r\mathbb{E}P} \le \frac{-\log(1 - 2\omega(c))}{2c}. \tag{3}$$

The embeddings are finite and exact, although their dimensions can be very large.

In fact, one fixed improvement on the trivial bound suffices: if there are $c > 0$ and $\eta < 1/2$ such that

$$\mathcal{E}(\sqrt{P}) \le \eta k \mathbb{E}P \quad \text{whenever } P \ge 0 \text{ and } \deg P \le ck,$$

then the same tensor-code argument proves (i). For any positive rational $c' < \min(c, 1/2)$ it gives $C = -\log(1 - 2\eta)/(2c')$. Here the trivial universal bound is $\mathcal{E}(\sqrt{P}) \le k\mathbb{E}P/2$, obtained by expanding each edge square and using $\sqrt{P(x)P(y)} \ge 0$.

4 A signed obstruction, including a global absolute margin

For a real function P define its signed square root

$$M(P) = \operatorname{sgn}(P)\sqrt{|P|}, \qquad \|M(P)\|_2^2 = \mathbb{E}|P|.$$

Theorem 3. *There is a sequence of real polynomials G_j on cubes of dimensions N_j, with degrees r_j, such that*

$$|G_j(x)| \ge 1 \quad \textit{for every } x, \qquad \frac{r_j}{N_j} \longrightarrow 0, \qquad \frac{\mathcal{E}(M(G_j))}{N_j \mathbb{E}|G_j|} \ge \frac{1 - e^{-1}}{4}.$$

Each G_j has all its nonzero Walsh coefficients on a single level. Consequently no universal bound $\mathcal{E}(M(P)) \le k\omega(\deg(P)/k)\mathbb{E}|P|$ with $\omega(t) \to 0$ can hold for arbitrary signed polynomials, even under the displayed global absolute-margin condition.

Proof. For odd m, set $S_m = \sum_{i=1}^m x_i$ and $a_m = \mathcal{E}(M(S_m))/\mathbb{E}|S_m|$. We first verify

$$a_m = \frac{1}{8}\log m + O(1). \tag{4}$$

Let T be a sum of $m - 1$ independent signs, and put $p_0 = \mathbb{P}(T = 0)$. Coordinate symmetry gives

$$\mathcal{E}(M(S_m)) = \frac{m}{4}\mathbb{E}\big(M(T+1) - M(T-1)\big)^2, \qquad \mathbb{E}|S_m| = mp_0.$$

The squared increment is 4 at $T = 0$. For positive even $t \geq 2$, it equals

$$d(t) = \frac{4}{(\sqrt{t+1} + \sqrt{t-1})^2} = \frac{1}{t} + O(t^{-3}),$$

and the expression is even in t. The local binomial estimate $\mathbb{P}(T = t)/p_0 = 1 + O((t^2+1)/m)$, uniformly for even $2 \leq t \leq \sqrt{m}$, yields

$$2 \sum_{\substack{2 \leq t \leq \sqrt{m} \\ t \text{ even}}} \frac{\mathbb{P}(T = t)}{p_0} d(t) = \frac{1}{2} \log m + O(1).$$

The local-estimate errors and the $O(t^{-3})$ terms are summable with total $O(1)$. For $t > \sqrt{m}$, use $d(t) \leq C/\sqrt{m}$ and $p_0 \asymp m^{-1/2}$ to bound the remaining normalized contribution by $O(1)$. Dividing the energy identity by mp_0 proves (4).

Fix odd m and form the ℓ-fold tensor product $F_\ell = \prod_{b=1}^{\ell} S_m(x^{(b)})$. It is homogeneous of Walsh degree ℓ, and $|F_\ell| \geq 1$. The signed square root tensorizes. If Z has the Walsh degree distribution of $M(S_m)/\sqrt{\mathbb{E}|S_m|}$, then $\mathbb{E}Z = a_m$, and the corresponding degree distribution for $M(F_\ell)$ is $Z_1 + \cdots + Z_\ell$.

Choose rational $\theta_\ell \in (0, 1/2)$ with $\ell\theta_\ell \to 1/(2a_m)$ and apply Lemma 2 to obtain $G_\ell = F_\ell \circ A$ on a K_ℓ-cube. The lift preserves $|G_\ell| \geq 1$, and (1) gives

$$\frac{\mathcal{E}(M(G_\ell))}{K_\ell \mathbb{E}|G_\ell|} = \frac{1}{2} \mathbb{E}\left[1 - (1 - 2\theta_\ell)^{Z_1 + \cdots + Z_\ell}\right] \longrightarrow \frac{1 - e^{-1}}{2}.$$

Here the law of large numbers and bounded convergence justify the limit. Homogeneity and the exact character-degree formula give

$$\frac{\deg G_\ell}{K_\ell} = \frac{1}{2}\big(1 - (1 - 2\theta_\ell)^\ell\big) \longrightarrow \frac{1}{2}(1 - e^{-1/a_m}) \leq \frac{1}{2a_m}.$$

For each increasing odd m, choose ℓ sufficiently large to make the energy ratio at least $(1 - e^{-1})/4$ and the degree fraction at most $1/a_m$. Equation (4) then gives the required diagonal sequence. All nonzero characters of each G_ℓ have the same physical degree, so the final homogeneity assertion also follows. □

The global margin in Theorem 3 is absolute. Indeed $\mathbb{E}|G_\ell| = (\mathbb{E}|S_m|)^\ell$ can be enormous, so the theorem does not assert a unit margin after normalization by $\mathbb{E}|G_\ell|$ or by a neural-network source cost. Nor is agreement with any specified target parity imposed. The result therefore obstructs a universal analytic shortcut, not the desired learning lower bound.

It also does not contradict a bound for the different nonlinear Rayleigh functional $\mathbb{E}\,\mathrm{sgn}(P)\Delta P$. For the homogeneous polynomials just constructed, this functional is exactly $\deg(P)\mathbb{E}|P|$, while the signed-square-root energy has the behavior in Theorem 3. Finally, $|S_m|$ itself does not have degree one; the signed counterexample makes no claim against the nonnegative-polynomial conjecture in Theorem 1.

5 Fixed parity accuracy does not remove the signed obstruction

The preceding counterexamples need not predict parity. A selector construction adds arbitrary fixed accuracy while retaining the energy obstruction. The absence of a source-cost bound remains essential to the scope of the conclusion.

Corollary 4. *Fix $\alpha < 1$. There is a sequence of real polynomials P_j on n_j-cubes such that*

$$|P_j| \geq 1 \text{ everywhere}, \qquad \frac{\deg P_j}{n_j} \longrightarrow 0, \qquad \mathbb{P}\left(\mathrm{sgn}(P_j) = \prod_{i=1}^{n_j} x_i\right) \geq \alpha,$$

and

$$\frac{\mathcal{E}(M(P_j))}{n_j \mathbb{E}|P_j|} \geq \frac{1 - e^{-1}}{16}.$$

No bound on actual neural-network source cost or posterior cell energy is asserted.

Proof. Let B on a K-cube be a member of the sequence constructed in Theorem 3, and put $b_0 = (1 - e^{-1})/4$. Its code has zero columns by Lemma 2, so B ignores at least one physical bit. Hence $\operatorname{sgn} B$ agrees with full parity f_K on exactly half of the cube.

For any fixed $\varepsilon > 0$ there is a polynomial G of degree $O_\varepsilon(\sqrt{K})$ with $|G| \geq 1$ everywhere and $\mathbb{P}(\operatorname{sgn} G = f_K) \geq 1-\varepsilon$. To see this, let $h(y) = (K - \sum_i y_i)/2$ be Hamming weight. Choose integers $a \leq b$ with $\mathbb{P}(a \leq h \leq b) \geq 1-\varepsilon$ and $b - a \leq \sqrt{K/\varepsilon} + 2$, using Chebyshev's inequality and rounding. The polynomial

$$G_0(h) = (-1)^b \prod_{j=a}^{b-1} (h - j - 1/2)$$

has sign $(-1)^h$ for integer $a \leq h \leq b$, and has no zeros at any integer. Divide by its positive minimum absolute value on $\{0, \ldots, K\}$ to obtain G.

Add t selector bits z, where t is fixed, and write $H(z) = \prod_{i=1}^{t}(1 + z_i)/2$, $\chi(z) = \prod_{i=1}^{t} z_i$, and $p = 2^{-t}$. For $A \geq 1$ set

$$P(z, y) = H(z)AB(y) + (1 - H(z))\chi(z)G(y).$$

This has global absolute margin one, degree at most $t + \max(\deg B, \deg G) = o(K)$, and full-parity accuracy at least

$$p/2 + (1 - p)(1 - \varepsilon).$$

Choose the fixed t and ε so this is at least α. The old-coordinate energies and nonnegativity of selector-coordinate energies give

$$\mathcal{E}(M(P)) \geq pA\mathcal{E}(M(B)), \qquad \mathbb{E}|P| = pA\mathbb{E}|B| + (1 - p)\mathbb{E}|G|.$$

Choose $A \geq (1 - p)\mathbb{E}|G|/(p\mathbb{E}|B|)$. For $K \geq t$ this yields

$$\frac{\mathcal{E}(M(P))}{(K + t)\mathbb{E}|P|} \geq \frac{b_0 K}{2(K + t)} \geq \frac{b_0}{4}.$$

Taking the sequence of B's proves the corollary. □

The multiplier A separates prediction accuracy from normalized signed-root energy. A separate direct calculation in the motivating posterior cell model shows that this particular selector family has diverging actual repair cost, even when all radial sign relays are allowed. That calculation uses wrong-to-correct neighbor contrasts; large amplitude alone would not justify its conclusion. The cell model and that additional argument are not needed for the analytic results proved here. Thus these counterexamples are excluded from a uniformly bounded-cost calibration setting, and the desired calibration statement for accurate heads with bounded actual source cost remains open. The natural-scale learning target remains open as well.